Piyush Singhal

+91-9694984312 | piyush.singhal.2004@gmail.com | linkedin/piyush/ | Coding Profile | Github Profile | Personal Portfolio

PROFESSIONAL SUMMARY

Final-year Computer Science student who builds **full stack** web apps with **Next.js**, **TypeScript** and **PostgreSQL**, and **LLM** features with **RAG** and **REST APIs**. Comfortable with **data structures and algorithms**, and experienced in **debugging**, **Docker** and **automated testing** from a SWE internship.

EDUCATION

NIMS University, Jaipur, India — **CGPA: 8.90/10** — **2023 – 2027**
Bachelor of Technology in Computer Science Engineering

Step By Step Sr. Sec. School — **Percentage: 90%** — **2022**
CBSE (Class XII)

Step By Step Sr. Sec. School — **Percentage: 86.2%** — **2020**
CBSE (Class X)

TECHNICAL SKILLS

Languages:	C, C++, Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
Core CS:	Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Web & Backend:	React.js, Next.js, Node.js, Express.js, REST APIs, PostgreSQL, Supabase, Celery, Redis
AI/ML:	Machine Learning, LLMs, RAG, LangChain, NLP, Embeddings, Semantic Search, Vector Search (FAISS), LLM Evaluation, PyTorch
Tools & DevOps:	Git, GitHub, CI/CD, Docker, Automated Testing, AWS, Vercel, Streamlit

EXPERIENCE

SWE Intern — **July 2026 – August 2026**
Ambiguity Labs (Remote)

- Created **20+ engineering tasks** from real repository issues covering **debugging**, API failures, migrations, feature development and infrastructure changes.
- Reproduced bugs in existing codebases, including **TypeScript** errors, **C/C++** memory failures, workflow and config issues.
- **Dockerized** task environments; wrote reference solutions and **automated testing** (oracle checks) to verify behavior.
- Evaluated **Claude** and **GPT-based coding agents**, analyzing task completion, correctness and failure patterns.

Open Source Contributor — **Jan 2026 – Apr 2026**
AOSSIE – EduAid Project (Remote)

- Made **LLM-based quiz generation** more reliable with deterministic output, response validation, quality scoring and **long-document** support.
- Improved **backend efficiency** by trimming API response payloads and adding compression, caching, batching and **lazy model loading**.
- Used **Celery** and **Redis** for **asynchronous AI inference** and background task processing.
- Fixed API and quiz bugs, and added input validation, security fixes, and quiz regeneration, review and sharing.

PROJECTS

Personal Finance Manager Platform | GitHub Link | Website Link — **Skills:** *Next.js, TypeScript, PostgreSQL*

- Implemented **user authentication** and per-user tracking of **income, expenses, budgets and transactions**.
- Added **savings-goal** and **bill-reminder** features; **deployed** the app on **Vercel**.

PDF Chat Application | GitHub Link | Website Link — **Skills:** *LangChain, FAISS, PyPDF2, Streamlit*

- Developed a **RAG pipeline** that extracts PDF text, chunks it and generates **embeddings** for **semantic search**.
- Indexed **20+ PDFs** in **FAISS**, achieving about **83% accuracy** on evaluated queries.
- Used **LangChain** to retrieve relevant context for natural-language question answering.

Smart Study Management Platform | GitHub Link | Website Link — **Skills:** *Next.js, TypeScript, PostgreSQL*

- Built **personalized study plan and timetable** generation to organize daily learning and academic tasks.
- Centralized tasks, resources and progress tracking in one place; **deployed** on **Vercel**.

ACHIEVEMENTS

- Secured **AIR 83** among **20,000+** participants in **CodeChef Starters 184** (2025).
- Rated **1750+** on CodeChef, **1750+** on LeetCode and **1400+** on Codeforces.
- Solved **500+ DSA problems** across CodeChef and LeetCode.

CERTIFICATIONS

- OpenAI Academy X NxtWave Buildathon
- Deloitte Australia – Technology Simulation
- Walmart USA – Advanced SWE Job Simulation
- Responsive Web Design Certification